

66. PQRS is a parallelogram. If $\overrightarrow{PR} = \vec{a}$ and $\overrightarrow{QS} = \vec{b}$, then what is $\overrightarrow{PQ}$ equal to?

- (a) $\vec{a} + \vec{b}$
- (b) $\vec{a} - \vec{b}$
- (c) $\frac{\vec{a} + \vec{b}}{2}$
- (d) $\frac{\vec{a} - \vec{b}}{2}$

67. Let $\vec{a}$ and $\vec{b}$ are two unit vectors such that $\vec{a} + 2\vec{b}$ and $5\vec{a} - 4\vec{b}$ are perpendicular. What is the angle between $\vec{a}$ and $\vec{b}$?

- (a) $\frac{\pi}{6}$
- (b) $\frac{\pi}{4}$
- (c) $\frac{\pi}{3}$
- (d) $\frac{\pi}{2}$

68. Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ be unit vectors lying on the same plane. What is

$$\left\{ \left(3\vec{a} + 2\vec{b} \right) \times \left(5\vec{a} - 4\vec{c} \right) \right\} \cdot \left(\vec{b} + 2\vec{c} \right)$$

equal to?

- (a) -8
- (b) -32
- (c) 8
- (d) 0

69. What are the values of x for which the angle between the vectors $2x^2\hat{i} + 3x\hat{j} + \hat{k}$ and $\hat{i} - 2\hat{j} + x^2\hat{k}$ is obtuse?

- (a) $0 < x < 2$
- (b) $x < 0$
- (c) $x > 2$
- (d) $0 \leq x \leq 2$

70. The position vectors of vertices A, B and C of triangle ABC are respectively $\hat{j} + \hat{k}$, $3\hat{i} + \hat{j} + 5\hat{k}$ and $3\hat{j} + 3\hat{k}$. What is angle C equal to?

- (a) $\frac{\pi}{6}$
- (b) $\frac{\pi}{4}$
- (c) $\frac{\pi}{3}$
- (d) $\frac{\pi}{2}$

71. Let $z = [y]$ and $y = [x] - x$, where $[.]$ is the greatest integer function. If x is not an integer but positive, then what is the value of z ?

- (a) -1
- (b) 0
- (c) 1
- (d) 2

72. यदि $f(x) = 4x + 1$ और $g(x) = kx + 2$ इस प्रकार है कि $f \circ g(x) = g \circ f(x)$ है, तो k का मान क्या है ?

- (a) 7
- (b) 5
- (c) 4
- (d) 3

73. फलन $f(x) = \log_{10}(x^2 + 2x + 11)$ का न्यूनतम मान क्या है ?

- (a) 0
- (b) 1
- (c) 2
- (d) 10

74. $\int (x^x)^2 (1 + \ln x) dx$ किसके बराबर है ?

- (a) $x^{2x} + c$
- (b) $\frac{1}{2} x^{2x} + c$
- (c) $2x^{2x} + c$
- (d) $\frac{1}{2} x^x + c$

75. $\int e^x \{1 + \ln x + x \ln x\} dx$ किसके बराबर है ?

- (a) $xe^x \ln x + c$
- (b) $x^2 e^x \ln x + c$
- (c) $x + e^x \ln x + c$
- (d) $xe^x + \ln x + c$

76. $\int \frac{(\cos x)^{1.5} - (\sin x)^{1.5}}{\sqrt{\sin x \cdot \cos x}} dx$ क्या है ?

- (a) $\sqrt{\sin x} - \sqrt{\cos x} + c$
- (b) $\sqrt{\sin x} + \sqrt{\cos x} + c$
- (c) $2\sqrt{\sin x} + 2\sqrt{\cos x} + c$
- (d) $\frac{1}{2}\sqrt{\sin x} + \frac{1}{2}\sqrt{\cos x} + c$

77. यदि $y = \frac{x\sqrt{x^2 - 16}}{2} - 8 \ln |x + \sqrt{x^2 - 16}|$ है, तो $\frac{dy}{dx}$ किसके बराबर है ?

- (a) $x\sqrt{x^2 - 16}$
- (b) $x - \sqrt{x^2 - 16}$
- (c) $\sqrt{x^2 - 16}$
- (d) $4\sqrt{x^2 - 16}$

78. यदि $y = (x^x)^x$ है, तो निम्नलिखित में से कौन-सा कथन सही है ?

- (a) $\frac{dy}{dx} + xy(1 + 2\ln x) = 0$
- (b) $\frac{dy}{dx} - xy(1 + 2\ln x) = 0$
- (c) $\frac{dy}{dx} - 2xy(1 + \ln x) = 0$
- (d) $\frac{dy}{dx} + 2xy(1 + \ln x) = 0$